

Wellness CA - API Endpoint Reference

Server: 152.42.181.66:8000 | Auth: JWT Bearer + X-API-Token | JSON: camelCase

10 endpoints. All JSON uses camelCase. Auth: X-API-Token header + Bearer JWT. JWT from POST /login expires in 24 hours.

POST

/register (Xia Zihang)

Create new user account. Password bcrypt-hashed.

```
{"username":"alice","password":"pass123"}
-> 201 {"message":"Registered","userId":1}
```

POST

/login (Xia Zihang)

Authenticate. Returns signed JWT access token.

```
{"username":"alice","password":"pass123"}
-> 200 {"accessToken":"eyJ...", "tokenType":"bearer"}
```

GET

/records (Xia Zihang)

List current user's wellness records, newest first.

```
-> 200 [{"id":1,"sleepHours":7.5,"exerciseActivity":"Running","exerciseDuration":30,"recordDate":"2025-07-01","notes":...}]
```

POST

/records (Xia Zihang)

Create a new wellness record.

```
{"sleepHours":7.5,"exerciseActivity":"Running","exerciseDuration":30,"recordDate":"2025-07-01","notes":"..."}
-> 201 {"message":"Created","id":1}
```

PUT

/records/{id} (Xia Zihang)

Update a record (own records only).

```
Same body as POST -> 200 {"message":"Updated"}
```

DELETE

/records/{id} (Xia Zihang)

Delete a record (own records only).

```
No body -> 200 {"message":"Deleted"}
```

POST

/chat (Huang Qianer)

AI chat with RAG enhancement. Retrieves relevant wellness context from ChromaDB before calling DeepSeek.

```
{"message":"Why am I tired?"}
-> 200 {"reply":"Based on your records..."}
```

GET

/recommendations (Xia Zihang)

Simple AI tips from last 7 days of wellness data.

```
-> 200 {"recommendations":["Go to bed earlier","..."]}
```

POST

/agent/recommend (Cai Peilin)

Agentic AI: plan-and-execute loop with function calling. Analyzes sleep/exercise trends and saves evidence-grounded recommendation.

```
No body -> 200 {"recommendation":"...", "evidence":[...], "iterations":3, "savedId":42}
```

GET

/agent/recommend/history (Cai Peilin)

List past agent-generated recommendations.

```
-> 200 [{"id":1,"content":"...", "evidence":[...], "iterations":3, "createdAt":"..."}]
```

Authentication Headers

```
Authorization: Bearer <jwtAccessToken>
X-API-Token: team-wellness-2025
Content-Type: application/json
```

JWT obtained from POST /login. Android stores in SharedPreferences. Redirect to login on 401. X-API-Token is team-shared gateway secret.

Sidecar Services (internal, not exposed to Android)

RAG (Huang Qianer) :8001 - ChromaDB vector store + OpenRouter embeddings

Agent (Cai Peilin) :8002 - DeepSeek function-calling agent loop

Spring Boot proxies /agent/* to :8002; ChatService calls :8001 for RAG context.